

WICII ZHANG

STAFF EMBEDDED SOFTWARE ENGINEER · AI-NATIVE

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HammerAgent owner

#1 Modem App Hub

10K+ parser usage

5G RF software

AI triage

TECHNICAL SKILLS

AI / Agent Engineering

QGenie SDKLLM/NLPRAGEmbeddingsMCPA2AEvidence extractionHITL validation

AI System Architecture

FastAPIPipeline runnerSerial queueCrash recoverySQLiteThreshold engineTelemetryLearning loop

Wireless / RF Front-End

5G NR RF SWTxAGCDSDA/SDS TxSULULCASRS switchingPA/PAPM/XSW/ASMRFFE timing

Modem / Tools

PythonCGitHexagonARM IUMATLABQXDM/QUTSQTRACE

RF Hardware

PAADSAltiumCallboxVNASpectrum analyzerSignal source

EDUCATION

M.E., Electronic Science and Technology

Beijing University of Posts and Telecommunications · Jun 2019

Rank: 20 / 275

B.S., Communication Engineering

Liaoning Technical University · Jun 2016

Rank: 5 / 120 · CET-6: 539 / 710 · NCRE Grade 2

ACADEMIC PROJECTS

863 Program: Designed high-efficiency GaN Doherty power amplifiers covering LTE bands in China, targeting 52 dBm peak output power and over 40% drain efficiency at 6 dB output power back-off.

973 Program: Explored efficiency improvement methodology for base stations and implemented a novel power amplifier with efficiency above 55% at 8–12 dB OBO under a 150 MHz modulated signal.

AWARDS

2nd National Prize, China Post-Graduate Mathematical Contest in Modeling.

1st Prize Academic Scholarship of BUPT, three times.

2nd National Prize, China Undergraduate Mathematical Contest in Modeling.

National Scholarship of LNTU; 1st Prize Academic Scholarship of LNTU, four times.

1st Provincial Prize, Chinese Mathematics Competitions.

LANGUAGES

English — Professional working proficiency (CET-6: 539/710)

Mandarin Chinese — Native

PROFESSIONAL SUMMARY

- Owner of HammerAgent, an AI-assisted modem debugging system built on QGenie SDK/model integration, agentic tool workflows, domain-specific RAG, evidence score, and rule-driven log analysis for RF, Digital, and MAC UL debug.
- Architected Hammer/HammerAgent as an E2E triage platform with FastAPI web portal/orchestrator, local pipeline runner, parser/analysis engine, cross-layer correlation, deterministic verdict gate, advisory AI layer, feedback store, and continuous learning loop.
- Staff RF Embedded Software Engineer with deep experience in 5G modem RF software, Tx architecture, RFFE control, product bring-up, customer escalation, and Hexagon & ARM IU processor development.

WORK EXPERIENCE

Qualcomm — Staff RF Software Engineer

Dec 2024 – Present · Pudong, Shanghai

- Created HammerAgent, an end-to-end AI-assisted debugging agent for RF, Digital, and MAC UL analysis, integrating QGenie SDK/model access, in-house embeddings, skill/tool orchestration, evidence extraction, and human-in-the-loop diagnosis.
- Co-architected Autologparser from parser automation into a measurable closed-loop triage architecture with QUTS/QXDM parsing, opcode registry, QTRACE/IQ capture, SQLite artifacts, FastAPI orchestration, dashboard telemetry, feedback ratings, and email/report delivery.
- Drove early-stage design and technical proposals with RF System and Modem System teams for TxAGC evolution, uplink power limiting, envelope expansion, and UL enhancement from concept to bring-up readiness.
- Designed and debugged RFFE control behavior across PA, PAPM, XSW, ASM, and tuner paths; served as point of contact for DSDA Tx and Intra-CA Tx feature design, triage, and customer support.

Qualcomm — Senior Software Engineer

May 2019 – Nov 2024 · Pudong, Shanghai

- Developed and brought up 5G modem RF software in highly concurrent embedded environments, covering RF/BB integration, Tx behavior, RFFE control, modem timing, certification, performance, stability, and customer issue resolution.
- Supported onsite and remote factory bring-up for Apple 5G products, including modem integration, RF calibration, production issue debugging, and cross-layer troubleshooting across RF software, modem control, and system behavior.
- Collaborated with System, RF, BB, AS, CE, and global development teams under aggressive schedules; mentored team members on RF debugging methodology, trace interpretation, and feature bring-up flow.

Apple R&D (Beijing) Limited — Wireless Design Innovator Intern

Jun 2018 – Apr 2019

- Built MMW RADAR applications with AWR1642 development kits.
- Explored intelligent issue triage powered by NLP log analysis.
- Participated in wireless product innovation activities with project managers, innovators, design engineers, manufacturing teams, and third-party testing labs.

Ericsson — Product Integration Intern

Jun 2018 – Jul 2018

- Responsible for RRU HW integration of Ericsson radio products, troubleshooting complex problems from product development, manufacturing, and field customer escalation.
- Core contributor to product quality assurance for radio hardware throughout the product lifecycle.

Keysight Technologies — Application Engineer Intern

Sep 2017 – May 2018

- Delivered solutions and guided customers on Keysight test equipment and services at the system/solutions level.
- Integrated Keysight products into customer environments, bridging test & measurement with real-world deployment scenarios.

PUBLICATIONS & PATENTS

- C. Yu, W. Zhang and Y. Liu, "A broadband Doherty power amplifier based on rotated T-shape load modulation network," 2018 IEEE MTT-S International Wireless Symposium, Chengdu, China, 2018.
- 2505340WO1 — Power allocation handling for uplink component carriers in an uplink carrier aggregation configuration.
- WO2022222023 — UE capability information for frequency band combination conflict.
- WO2022252169A1 — Dynamic code block group allocation.